

SCALING

MOOSIC

WITH MACHINE LEARNING



PRESENTED BY: VASIA SPYROPOULOU

CAN WE **AUTOMATE** PLAYLIST CREATION?



YES!

**5,235 SONGS →
27 PLAYLISTS OF
152-213 SONGS EACH**



HOW WE DID IT: A 2-STEP APPROACH

5235 SPOTIFY SONGS

9 AUDIO FEATURES:

- DANCEABILITY
- ENERGY
- VALENCE
- ACOUSTICNESS
- LOUDNESS
- SPEECHINESS
- INSTRUMENTALNESS
- LIVENESS
- TEMPO

**UNSUPERVISED ML
(K-MEANS CLUSTERING)**

→ **8 BROAD GROUPS**

**SPLIT BY MOOD
SCORE**

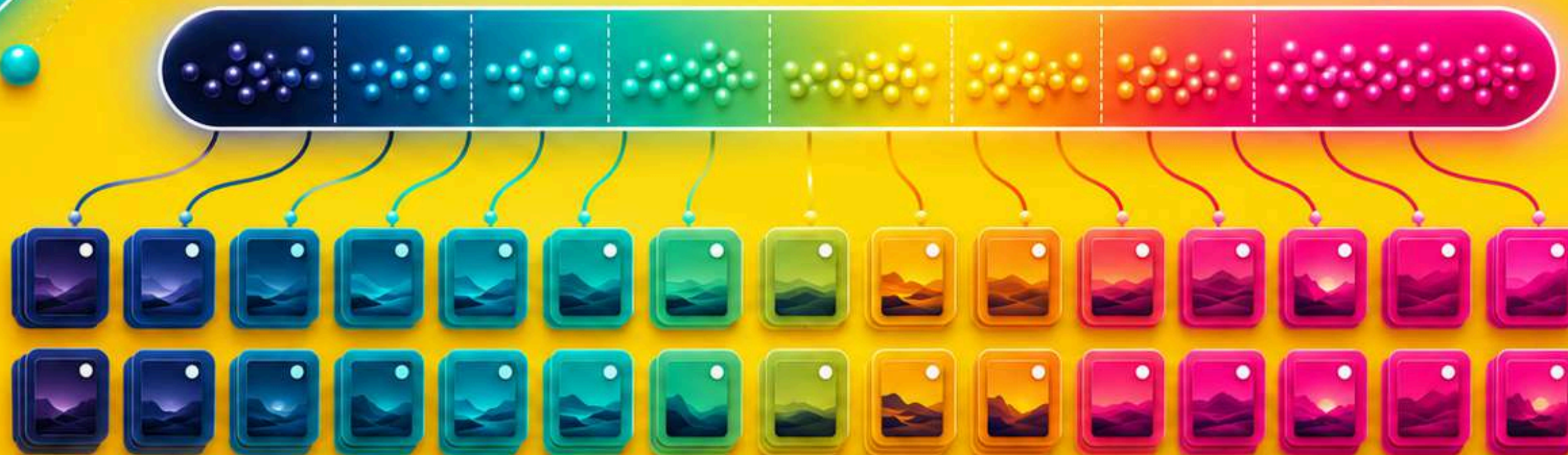
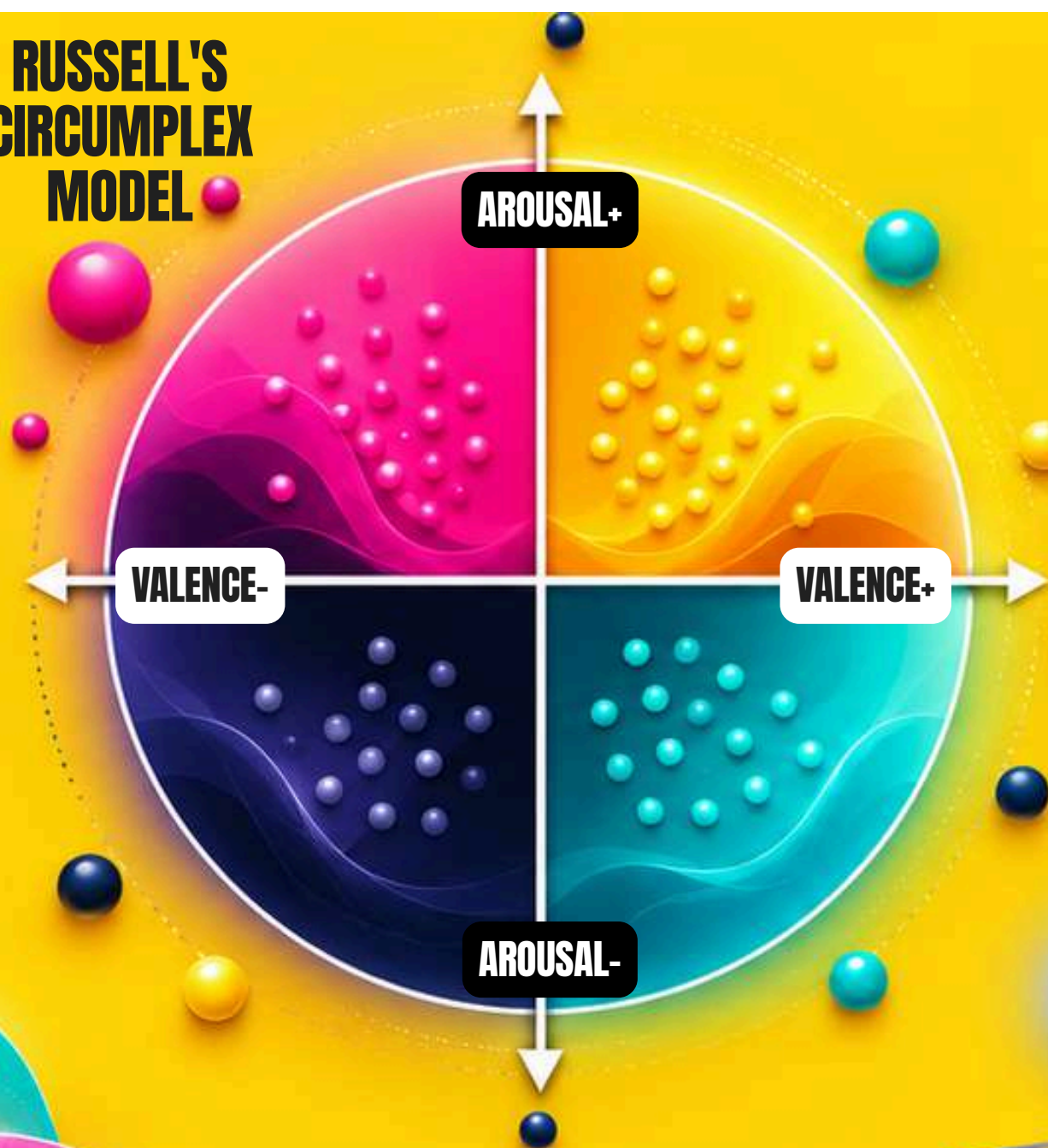
→ **27 PLAYLISTS
(152-213 SONGS)**



SPLITTING BY MOOD



**RUSSELL'S
CIRCUMPLEX
MODEL**





PLAYLIST EXAMPLE A

PLAYLIST 1_1 (193 SONGS)

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- **MAGALENHA - SÉRGIO MENDES**
- **ALEGRIA - BATIDA**
- **LA VUELTA AL MUNDO - CALLE 13**
- **DUSTY MEN - SAULE**
- **HEART OF GLASS - SPECIAL MIX - BLONDIE**
- **AVANT GARDENER - COURTNEY BARNETT**
- **PAELLA - DA SOUZA**
- **I MISS YOUR BONES - HOSPITALITY**
- **CONSECUTIVE SEASONS - TWERPS**
- **CENTRO DI GRAVITÀ PERMANENTE - 2004 - REMASTER; - FRANCO BATTIATO**
- **....AND MORE**

PLAYLIST EXAMPLE **B** + HONEST ASSESSMENT



PLAYLIST 0_0 (197 SONGS)

- =====
- **MOSTLY: CLASSICAL, AMBIENT, AND NEO-CLASSICAL TOGETHER.**

BUT

- **THE QUEST OF ABSURDITY - ASPHYX**
- **CURSED - MORGOOTH**



But the algorithm also placed Asphyx and Morgoth, two death metal bands in a playlist full of Chopin and Brian Eno.

Same low valence, very different sound.

This shows exactly why we need genre data.

DO AUDIO FEATURES **WORK?**



CAPTURE:

ENERGY, MOOD, ACOUSTIC VS ELECTRONIC



MISS:

GENRE, LYRICS LANGUAGE, CULTURAL CONTEXT



WOULD HELP:

GENRE LABELS, LYRICS ANALYSIS, USER BEHAVIOR

NEXT STEPS

- **A/B TEST: ML PLAYLISTS VS EXPERT-CURATED**
- **ADD GENRE LABELS + LYRICS ANALYSIS**
- **USER FEEDBACK LOOP**
- **ALSO EXPLORE SUPERVISED MACHINE LEARNING**
- **"OUR RECOMMENDATION: ML AS STARTING POINT, EXPERTS AS FINISHING TOUCH"**
- **"QUESTIONS?"**

THANK YOU



SOURCES: Valence as a measure of happiness

An emotion analysis algorithm in identifying moods of song and valence through musical features